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**Single cell RNA-sequencing captures
phenotypic changes in the repertoire of
cardiac stromal cells following ischemic injury**

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requirements for the MSc degree in Genes, Drugs and Stem cells –
Novel Therapies

Statement of Originality

I certify that this thesis, and the research to which it refers, are the product of my own work, conducted during the current year of the MSc in Genes, Drugs and Stem cells - Novel Therapies at Imperial College London. Any ideas or quotations from the work of other people, published or otherwise, or from my own previous work are fully acknowledged in accordance with the standard referencing practices of the discipline.

Dataset 1: scRNA-seq data of PDGFR α ⁺ fated cells from unoperated murine hearts and hearts 1-, 7- and 30- days post-MI were kindly provided by the Nosedá lab. Breeding and maintenance of mice was carried out by Dr. Chrystalla Constantinou. Surgical induction of MI was performed by Dr. Mohamed Bellahcene. Single cell preparation, generation of 10X GEMs and preparation of 10X libraries was performed by Ricardo Privolizzi, Dr. Chrystalla Constantinou (unoperated hearts) and Sara Samari (1-, 7- and 30- days post-MI hearts). Sequencing of unoperated hearts was performed by the Wellcome Sanger Institute. Sequencing of the post-MI hearts was performed by the BRC Genomics Facility at Imperial College London.

Dataset 2: scRNA-seq data of mouse cardiac interstitial cells from sham-operated hearts and hearts 3- and 7- days post-MI were publicly available from the ArrayExpress database at EMBL-EBI under accession code E-MTAB-7376 (Farbehi et al., 2019).

Dataset 3: scRNA-seq data of the murine cardiac non-myocyte cell pool from unoperated hearts were publicly available from the ArrayExpress database at EMBL-EBI under accession code E-MTAB-6173 (Skelly et al., 2018).

Thank you to Dr. Antonio Manuel Almeida Miranda for processing the raw scRNA-seq datasets in Cell Ranger and performing initial clustering to generate cell type labels.

Abstract

Myocardial infarction (MI) is a leading cause of death worldwide. Cardiac fibroblasts play a leading role in wound healing in response to MI following cardiomyocyte death, by promoting maladaptive remodelling which contributes to loss of cardiac function. Reperfusion therapies, currently the first choice of treatment, restore blood flow but are insufficient to restore the lost cardiac muscle tissue and function. A better understanding of the cardiac fibroblast cell landscape is crucial for the development of strategies to induce cardiac repair. Single-cell RNA-sequencing (scRNA-seq) provides a new tool to explore the cardiac stromal compartment and understand the cellular drivers of adverse cardiac remodelling with higher resolution and precision. Here, we computationally analysed a mix of newly generated and publicly available scRNA-seq data from unoperated, sham-operated, and operated mouse hearts collected at various timepoints following experimental MI. We used low resolution Leiden clustering to identify and subsequently annotate 10 major cell types within the whole dataset comprising of 75,145 cells. Further sub-clustering identified 7 subpopulations within the cardiac stromal fibroblast compartment which comprised of 42,181 cells, and 12 subpopulations within the macrophage compartment. Within fibroblasts, specificity of *Wif1* expression was confirmed globally. Interactions were computationally predicted between fibroblasts and macrophages using *CellPhoneDB*, identifying a key receptor-ligand pair CD74/MIF which might be a promising target to identify cardioprotective mechanisms. Overall, this analysis has given a deeper insight into the heterogeneity of cardiac stromal fibroblast cells and their dynamic changes in gene expression and functions post-MI.

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Abbreviations

BBKNN (batch-balanced k-nearest neighbours)

CABG (coronary artery bypass graft)

cMSC (cardiac mesenchymal stromal cell)

CSC (cardiac stem cell)

ECM (extracellular matrix)

HF (heart failure)

MI (myocardial infarction)

MSC (mesenchymal stromal cell)

NGS (next-generation sequencing)

PCA (principle component analysis)

PCI (percutaneous coronary intervention)

PDGFR α (Platelet Derived Growth Factor Receptor Alpha)

QC (quality control)

SCCAF (single cell clustering assessment framework)

scRNA-seq (single-cell RNA-sequencing)

SP (side-population)

UMAP (uniform manifold approximation projection)

Introduction

Epidemiology of MI

Despite tremendous advances brought about by reperfusion therapies, myocardial infarction (MI) remains a leading cause of death and a severe daily burden on patients worldwide (Asaria *et al.*, 2017). MI occurs when a region of the heart is deprived of oxygen due to diminished blood flow to one of the coronary arteries, causing ischemic injury. On a cellular and molecular level, MI is characterised by the death of cardiomyocytes followed by an inflammatory response, activation of cardiac fibroblasts with abnormal production of extracellular matrix (ECM) proteins, and increased fibrosis (Frangogiannis, 2019; Berk *et al.*, 2007; Kong *et al.*, 2013). The subsequent maladaptive remodelling with a lack of cardiomyocyte replacement eventually leads to heart failure (HF).

Currently, treatments for MI rely largely on reperfusion via approaches including thrombolytic therapy, percutaneous coronary intervention (PCI) and coronary artery bypass graft (CABG) surgery. Reperfusion therapies can restore blood flow and dramatically increase immediate patient survival, but they cannot replace the lost muscle tissue nor induce its repair (Verma *et al.*, 2002). Furthermore, every 5% increase in infarct size leads to a 20% increase in the probability of developing HF in patients that have recently undergone PCI, highlighting the need to restrict ischemic damage as early as possible (Stone *et al.*, 2016).

Putative cardiac stem/progenitor cells

Cardiac stem/progenitor cells have been proposed to replace the lost cardiomyocytes after injury largely as a cell therapeutic product. Stem cells have a unique defining ability to self-renew and generate multiple cell lineages and are

therefore able to replace lost cells in a tissue or organ (Lee, 2018). The concept of potential resident adult stem cells in the heart was introduced after resident adult stem cells were observed in other systems and tissues, including the hematopoietic system, hair follicle and epidermis, striated muscle and gut/lung epithelium, as reviewed recently (Clevers & Watt, 2018). Numerous approaches have been suggested to characterise the dormant adult cardiac progenitors, including the use of expression of orphan receptor stem cell antigen-1 (Sca1) (Matsuura *et al.*, 2004; Oh *et al.*, 2003), receptor tyrosine kinase (c-kit) (Ellison *et al.*, 2013), and the side population (SP) dye-efflux phenotype (Oyama *et al.*, 2007; Nosedá *et al.*, 2015). In mouse MI models, isolated cardiac cells expressing the Sca1 surface molecule have exhibited high telomerase activity and bone morphogenic protein (BMP) pathway-dependent survival and differentiation *in vitro*, as well as post-MI migration to the myocardium and contribution towards cardiomyocyte turnover after therapeutic grafting *in vivo* (Oh *et al.*, 2003; Uchida *et al.*, 2013).

In line with these promising results, a range of investigative stem cell-based therapies have been tested in both preclinical and clinical studies, in hopes that they will rescue cardiomyocyte abundance in the adult mammalian MI heart. Such cellular products include heterogeneous bone marrow cells (BMCs), purified mesenchymal stromal cells (MSCs), cardiomyocytes derived from pluripotent stem cells (induced and embryonic), cardiopoietic cells derived from adult stem cells, heterogeneous cardiosphere-derived cells and purified cardiac stem and progenitor cells previously mentioned, that can be grafted by multiple modes of delivery (Banerjee *et al.*, 2018). Whilst the safety of different cell therapies including MSC- (Lalu *et al.*, 2018) and CSC-based (Fernández-Avilés *et al.*, 2018) approaches has been generalised, their efficacy remains controversial (Menasché, 2020; Nakamura & Murray, 2019; Witman *et al.*,

2020; Chien *et al.*, 2019). Another common challenge among cell therapies that we must overcome, is the poor retention of injected cell suspensions in the targeted area (Mercola *et al.*, 2011; Hong *et al.*, 2014; Nosedà *et al.*, 2015). Implementation of cell therapy approaches would require a definition of the best cellular candidates and optimization of biomaterials, such as scaffolds, to improve cardiac engraftment and retention and consequently the efficacy of such cell therapies (Higuchi *et al.*, 2017).

Nevertheless, the existence of an adult stem cell population in the heart has been challenged by the results of numerous fate-mapping studies using Cre/lox technology, which observed that both c-kit⁺/Sca1⁺ noncardiomyocyte cells (Senyo *et al.*, 2013; Vagnozzi *et al.*, 2018) or all noncardiomyocytes in the heart (Li *et al.*, 2018) do not differentiate into cardiomyocytes. Instead, Sca1⁺ noncardiomyocytes have been shown to generate fibroblasts and endothelial cells (Tang *et al.*, 2018), while resident cardiomyocytes appear to be the primary source of cardiomyocyte turnover post-MI (Senyo *et al.*, 2013). However, the irreversibility of MI implies that endogenous self-renewal of resident cardiomyocytes is insufficient to compensate for cells lost during injury, and furthermore, attempts to derive a “consensus” turnover rate of cardiomyocytes in adult mammalian hearts have remained elusive (Beltrami *et al.*, 2001; Doppler *et al.*, 2017; Bergmann *et al.*, 2009; Bergmann *et al.*, 2015).

As such, investigation into the progenitor potential of other cardiac cell types presents a promising research avenue for therapeutic strategies in MI. It remains unclear whether a true cardiac stem cell exists in the heart, or whether the stem progenitor phenotypes observed in previous experiments have been the result of functional changes occurring upon *in vitro* expansion. The differences in the molecular signature of the various endogenous cardiac progenitors described remain unclear, and the overlap of their signatures with cardiac fibroblasts (Chong *et al.*, 2011; Furtado

et al., 2014) and their functions is not fully understood. In fact, many features of cardiac fibroblasts are poorly characterised, and very little is known about their heterogeneity in the healthy hearts and how their molecular signature changes after MI.

Cardiac stromal fibroblast and macrophage responses post-MI

Previously, our team used fluorescence-activated cell sorting (FACS) and single-cell quantitative reverse transcription polymerase chain reaction (qRT-PCR) to define Sca1+PDGFRa+SP cells, a subset of cardiac stromal cells that have overlapping features with MSCs and fibroblasts (Nosedá *et al.*, 2015). Whilst cells with SP phenotype demonstrated high clonogenicity and expression of stem cell-related markers and cardiogenic markers *Gata4/6*, *Tbx5/20* and *Hand2* (Fig. 1. A), these were more specific to the PDGFRa+ cells, as Sca1+PDGFRa+non-SP cells shared these properties (Fig. 1. B). Sca1+ cells alone lacked clonogenicity, but SP cells with co-expression of Sca1 PDGFRa resembled MSC-like cardiac fibroblast colony-forming units (cCFU-F), but also differed from them as cCFU-F do not have a cardiogenic signature (Chong *et al.*, 2011). They also demonstrated cardiac protection by intramyocardial injection of cloned SP cells following MI (Fig. 1. C) and identified their specific functions underpinning protection, including modulation of the fate of macrophages (unpublished data, obtained using same cMSC-conditioned media as in Constantinou *et al.*, 2020), which have a crucial dynamic role in response to injury. In short, after MI macrophages initially show a pro-inflammatory phenotype (M1 state) and then undergo polarisation to their M2 reparative phenotype (Ma, Mouton & Lindsey, 2017). Later, using SmartSeq2, they defined the molecular signature of SP, with high expression of *Dkk3*, *Scara3* and *1500015O10Rik* (unpublished). There is still an unmet need to characterise these cells, considering their lack of established

molecular markers. Therefore, it is important to investigate the roles of and changes in PDGFRa+ cells as well as the SP cells, and also whether these have a potential effect on, and interact with, macrophages like we see *in vitro*.

Single cell transcriptomics

A better understanding of the cardiac fibroblast cell landscape in response to MI is crucial for the development of strategies to induce cardiomyocyte remodelling and repair. Recent advances in next-generation sequencing (NGS) technologies, especially single cell transcriptomics, have provided deeper insight into the distribution, heterogeneity, and differential gene expression within cell populations in different organisms and tissues (Massaia *et al.*, 2018). Single-cell RNA-sequencing (scRNA-seq) analysis allows exploration of the prevalence, heterogeneity and co-expression of the genes expressed in single cells, enabling the study of rare or novel cell populations and phenotypic responses that would fail to be detected from analysis of bulk RNA-seq or pooled cells. Previous scRNA-seq analyses of murine cardiac cells include transcriptional profiling of 10,519 noncardiomyocytes in healthy hearts (Skelly *et al.*, 2018), arguably an insufficient number of cells. Another study has taken this a step further by using samples from hearts post-MI, allowing the identification of the dynamics in certain cell lineages specifically in response to cardiac injury (Farbehi *et al.*, 2019). Whilst this team used a PDGFRa-GFP+ reporter line, this might be considered as inferior to more advanced fate mapping techniques using a Cre-lox system that labels cell progeny as well as positive cells, enabling identification of the embryonic region from which the cells have developed.

Here, we computationally analysed a mix of scRNA-seq datasets, including newly generated data by our team following a fate mapping experiment, and 2 publicly

available datasets (Skelly *et al.*, 2018, Farbehi *et al.*, 2019), collectively from samples from unoperated and sham-operated murine hearts and murine hearts collected at 4 timepoints following experimental MI.

Analysis was carried out with the following aims:

1. To explore the heterogeneity of the cardiac fibroblast stromal compartment, with a focus on definition of fibroblast subpopulations and to make inferences in their functions.
2. To define changes in the cardiac stromal fibroblast compartment in healthy hearts vs. after coronary artery ligation, and their interactions with macrophages.
3. To explore the expression of previously defined SP markers in the cardiac stromal fibroblast compartment in healthy hearts vs. after coronary artery ligation and make inferences in the functions of cells expressing these markers.

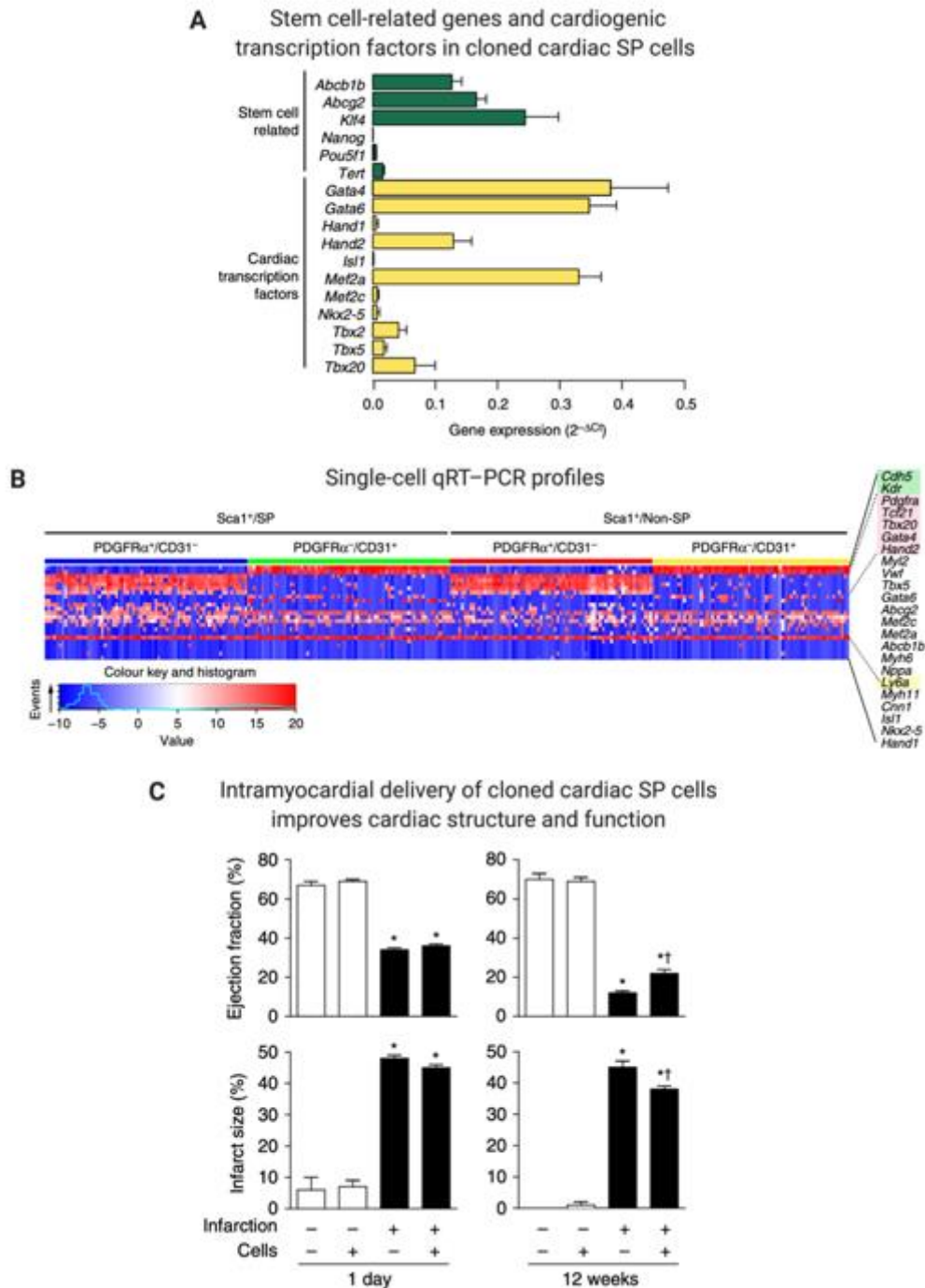


Figure 1. PDGFR α demarcates the cardiogenic clonogenic Sca1 $^{+}$ stem/progenitor cell in adult murine myocardium. A) Bar graph showing stem cell-related and cardiogenic gene expression in cloned cardiac SP cells. Bar graph mean $\pm$ s.e.m. (standard error of mean) for all 17 genes' expression in 20 independent clones. Genes are colour coded based on functional association or tissue specificity. **B)** Heatmap showing single-cell qRT-PCR profiles, showing co-expression of selected genes in Pdgfra $^{+}$ /CD31 $^{-}$ versus Pdgfra $^{-}$ /CD31 $^{+}$ cells from the SP and non-SP fractions. Expression is shown as ΔC_t values (blue, low; red, high). Genes are ordered based on the variance of mean expression levels of the four groups. Highlighted genes: green, Cdh5 and Kdr, enriched in PDGFR α +/CD31 $^{+}$ cells; light red, Pdgfra, Tcf21, Tbx20, Gata4 and Hand2, in PDGFR α +/CD31 $^{-}$ cells; yellow: Ly6a, in all four subpopulations. N=60–70 for each. **C)** Serial measurements showing improved ejection fraction and diminished infarct size at 12 weeks in infarcted hearts that received cloned cardiac SP cells. Bar graphs present the mean $\pm$ s.e.m. (sham+vehicle, n=4; sham+SP cells, n=10; MI+vehicle, n=8; MI+SP cells, n=20). *P<0.05 compared with the corresponding uninjured control hearts; † P<0.05 compared with infarction plus the cell-free vehicle control injection. (Figures were taken from Nosedá *et al.*, 2015).

Materials and Methods

Breeding and maintenance of mice

12-30 weeks old C57BL/6 double transgenic mice were used for immunostaining, with the addition of wild-type mice for generation of scRNA-seq data (Nosedá batch). Double transgenic mice were obtained by breeding PDGFR α GFP-Cre-ERT2/+ (Miwa & Era, 2015) and Rosa26tdTomato/+ (Madisen *et al.*, 2010) transgenic mice. For recombination, 500 μ g tamoxifen per gram of food was given to the mice for 5 consecutive days. At least 10g of food were given per day per mouse. Food was replaced daily. Normal powdered diet without tamoxifen was given after the end of tamoxifen diet for 2 consecutive days. For MI treated mice, MI was induced on the following day of the end of normal diet. One mouse was used per timepoint (timepoints consisted of Unop, D01, D07, D30; total of 4 mice).

Surgically induced myocardial infarction (MI)

To induce MI, coronary artery ligation was performed by Dr. Mohamed Bellahcene, as described previously by the team (Nosedá *et al.*, 2015; Fiedler *et al.*, 2019). For the 4 mice used, general anaesthesia was induced with 4% isoflurane briefly, then maintained at 2% in 100% O₂. 0.024 mg of buprenorphine was delivered to mice subcutaneously (i.e. average dose, 1.1 mg/kg; Vetergesic, Alstoe Animal Health, UK). The trachea of the mice were then intubated and ventilated mechanically with a tidal volume of 250 μ l and respiratory rate of 150 breaths/minute (Hugo-Sachs MiniVent type 845; Harvard Apparatus Ltd., Kent, UK). Heart rate, ECG and core body temperature were monitored throughout the surgery. The thorax was opened at the fourth intercostal space and then the pericardium was removed. The left anterior descending (LAD) was permanently ligated with a 6-0 polypropylene suture.

Myocardial ischemia was confirmed by blanching of the myocardium downstream of the suture and clear ST-segment elevation on ECG tracings. After surgery, mice were placed in a heated chamber for 20 minutes to rest. They were then moved to a normal holding cage with supplemental heat if necessary. For the unoperated mouse, treatment and surgery was not performed and here, hearts were collected. For MI-induced mice, hearts were collected at 1 day, 7 days and 30 days post-MI. For collection, general anaesthesia was induced again with 4% isoflurane briefly, then maintained at 2.5% in 100% O₂. Animals were subsequently perfused via intracardiac injection with PBS for 30 s and 4% PFA in PBS for 60 s prior to heart removal.

Single cell preparation

Mouse hearts were harvested, rinsed and cut in 1-2mm pieces before being placed in and gentleMACS C-tubes (Miltenyi Biotec) containing enzymatic digestion base solution (100 µgml⁻¹ Liberase TH Research grade and 50 µgml⁻¹ DNase I, HBSS 10 mM HEPES, 30 mM taurine) (Nosedá *et al.*, 2015). Mechanical and enzymatic digestion was performed at 37°C using gentleMACS Octo Dissociator (Miltenyi Biotec) with heaters. Cardiomyocyte-depleted single-cell suspensions were washed with base solution (HBSS 10 mM HEPES, 30 mM taurine) containing 20% fetal bovine serum (FBS, Gibco), filtered through 70 µm nylon strainer (BD Falcon), collected by centrifugation (330×g, 10 min, 4°C) and resuspended in base solution containing 0.2% FBS (Gibco). Cells were counted by Trypan blue and subsequently sorted by FACS (Nosedá *et al.*, 2015). Sorted cells are then Single cells are processed using Chromium Controller (10X Genomics) according to the manufacturer's protocol.

Chromium 10X library preparation

Cell suspension was adjusted to 1,000 cells/ μ l and loaded on the Chromium Controller (10X Genomics) with a targeted cell/nuclei recovery of 4,000 per reaction. 3' gene expression libraries were prepared according to the manufacturer's instructions of the v2 or v3 Chromium Single Cell Reagent Kits (10X Genomics). Quality control (QC) of cDNA and final libraries was done using Bioanalyzer High Sensitivity DNA Analysis or TapeStation System (both from Agilent). Libraries were sequenced using HiSeq 4000 (Illumina) with a minimum depth of 20,000 read pairs per cell.

Processing of 10X Genomics Chromium raw scRNA-seq data

Raw scRNA-seq data was processed using *Cell_Ranger* software (10X Genomics, version 3.1.0). FASTQ files were obtained from sequencing from the Illumina HiSeq platform (4 files). FASTQ files from the 2 sources of publicly available datasets were obtained from the ArrayExpress database (1 file, Skelly *et al.*, 2018; 3 files, Farbehi *et al.*, 2019). The *cellranger count* program used to create raw count matrices by aligning the 8 individual FASTQ files to a custom mouse reference genome with added tdTomato, GFP and STOP cassette sequences, generated using the *cellranger mkref* program. Details of custom genome: mouse GRCm38.p6 genome release, with tdTomato and STOP cassette from Ai9 plasmid (<https://www.addgene.org/22799/>), and EGFP-CreERT2 from pENTR eGFP-CreERT2 L1-R5 plasmid (<https://www.addgene.org/43961/>). For integration of the 8 count matrices, they were then imported into Python (version 3.6.8). An *anndata* object was created for each count matrix using the *AnnData* package (version 0.7.1) (Wolf *et al.*, 2018), and they were concatenated together using the *pandas* package (version 0.24.2) (McKinney,

2010) to produce one object for the whole dataset. Instead of considering each different sample/matrix, the lab of origin was assigned as the batch, as differences are expected between the different timepoints. After merging the datasets, the *Scrublet* package (version 0.2.1) was used to remove doublets by filtering out scrublet scores > 0.25 (Wolock *et al.*, 2018).

Pre-processing and analysis of raw data

Pre-processing and initial analysis was performed on the raw scRNA-seq count matrix containing the 3 concatenated datasets, using the *Scanpy* package (version 1.5.1) in Python (version 3.6.8). Cells were filtered as followed: $200 < n_genes < 3000$, $percent_mito < 40\%$, $n_cells > 50$ (n_genes : number of genes expressed per cell; $percent_mito$: percentage of mitochondrial genes per cell; n_cells : number of cells that express a gene). The expression matrix was total-count normalised to 10,000 reads per cell. A natural logarithm was applied to the normalized counts, with a pseudocount of 1 added for each gene. The effects of total counts per cell and percentage of mitochondrial genes were regressed out. A filter for genes with maximum standard deviation of 10 was applied, and the counts for each gene were scaled to unit variance. Principle component analysis (PCA) was then applied to the whole dataset (Pedregosa *et al.*, 2011) to generate a lower-dimensional representation of the complete gene space. The contribution of the top 50 principle components (PCs) towards the total gene expression variance was calculated by adding the explained variance percentages of these top PCs.

Clustering

Putative cell populations in the complete data set were manually annotated by Dr. Almeida Miranda, who used the 50-component PCA representation of the scaled transcript count data to generate a K-nearest neighbour graph, applied the Leiden clustering algorithm with multiple resolutions (Traag *et al.*, 2019), and used the expression values of well-known marker genes within the clusters to annotate different cell types. The *BBKNN* (batch-balanced K-nearest neighbours) package (version 1.3.6) was applied to generate the K-nearest neighbours graph (Polanski *et al.*, 2019). Within these manually annotated cell populations, separate PCAs were applied on the subsetting data to generate 50-component representations of the full gene expression space for each population. The *BBKNN* package was also applied on each population, followed by the Leiden clustering algorithm (Traag *et al.*, 2018), to sub-cluster the putative cell populations.

Clustering optimisation

Clustering of fibroblasts was optimised computationally by assessing the performance of the Leiden clustering algorithm to generate the most distinguishable subclusters at 3 different resolutions by changing the resolution parameter (0.2, 0.4, 1.1). Correlation matrix plots were computed with the Pearson's method and visualised using the *Scanpy* package (Wolf *et al.*, 2018). To generate receiver operator characteristic (ROC) curves, the *SCCAF* (single cell clustering assessment framework; Miao *et al.*, 2020) package (version 0.0.10) was used. Also, Calinski-Harabasz (CH; Calinski & Harabasz, 1974) and Silhouette scores (Rousseeuw, 1987) were computed using the *scikit-learn* package (version 0.22.2) over a range of

resolutions (0.2-1.9), on the PCA representation of the data for the top 50 PCs. These were visualised using the matplotlib package (version 3.0.3).

Differentially expressed genes (DEGs) identification

The *rank_genes_groups* function as part of the *Scanpy* package (version 1.5.1) was used to identify DEGs across groups. The logreg (logistic regression) method was used when identifying unique and group-specific makers as suggested (Ntranos *et al.*, 2018). The Wilcoxon (Wilcoxon rank-sum; Wilcoxon & Wilcox, 1964) method was used to identify further markers with significant differences in expression (filtered for the top 500 genes with lowest p-values, with q-value<1E-5 and log fold change>1).

Visualisation

After PCA and BBKNN, uniform manifold approximation projection (UMAP) was computed to further reduce dimensionality (McInnes *et al.*, 2018). Gene expression and sub-clustering results after Leiden clustering were visualised using the *umap* function (version 0.3.10) in the *Scanpy* package (version 1.5.1). Standard gene expression was visualised using the *Scanpy dotplot* function. DEGs calculation results for the logreg method were visualised using the *Scanpy rank_genes_groups_dotplot* function. A filter of minimum log fold change=2 was applied when plotting ranked genes across the main cell types. A filter of minimum log fold change=0.5 was applied when plotting ranked genes across timepoints within the F4 and F2 fibroblast subclusters. Cluster relationships after Leiden clustering were computed using hierarchical clustering and visualised using *Scanpy*.

Pathway/GO term enrichment analysis

The filtered DEGs calculated using the Wilcoxon method were imported into R Studio (version 3.6.0). Using script written by Dr. Michael Lee, the ToppGene Suite application ToppFun was used to identify the top biological processes (using the Gene Ontology NCBI Entrez Gene database) and pathways (using KEGG, BioCarta, BioCyc, Reactome, GenMAPP and MSigDB databases), which were visualised with heatmaps.

Computational detection of intercellular signalling

Each timepoint was subsetting from the whole dataset in Python (version 3.6.8). The *CellPhoneDB* package (version 2.1.4) was used on the individual files of the timepoints using default parameters (Efremova *et al.*, 2019). The significant means files were imported back into Python (version 3.6.8) and concatenated together into one anndata object. Interactions between fibroblasts and macrophages were subsetting.

Results

Pre-processing and analysis of raw data

To study the cardiac stromal compartment following MI, we used newly generated scRNA-seq data as well as publicly available datasets from healthy hearts and at various timepoints following MI. Datasets from 3 independent labs were used (Nosedá, unpublished data; Farbehi *et al.*, 2019; Skelly *et al.*, 2018) and the lab of origin was considered as the batch. The datasets from the Nosedá batch included timepoints: unoperated, day 1, day 7 and day 30 post-MI (total 4 hearts). The Harvey batch (Farbehi *et al.*, 2019) included timepoints: sham, day 3 and day 7 post-MI (total hearts unknown). The Skelly batch (Skelly *et al.*, 2018) included data from unoperated hearts only (total 4 hearts). Single cell transcriptomic analysis was performed using *Scanpy* (Wolf *et al.* 2018), a scalable Python-based pipeline that includes steps for pre-processing, visualization, clustering and differential expression testing. The pre-processing was performed on the raw scRNA-seq count matrix following the steps taken, highlighted in the schematic diagram shown (Fig. 2. A).

For quality control (QC), basic filtering was applied to single cells by removing those expressing less than 200 genes, as they are likely representing doublets or cell aggregates that were not removed with *Scrublet* (Wolock *et al.*, 2018). Cells expressing more than 3000 genes were also removed. Genes to be utilised in downstream analysis were also refined by excluding genes expressed in less than 50 single cells. Further filtering was applied to exclude cells that are stressed as indicated by a high percentage of mitochondrial genes. Here, a maximum threshold of 40% was used. This automatically filtered the number of total counts per cell, to 15,995. The violin plots for the 3 QC measures show the distribution of cells with the number of genes, number of total counts and percentage of mitochondrial before (Fig. 2. B) and

after (Fig. 2. C) QC filtering. The final dataset containing cells that passed QC included 75145 cells and 14624 genes in total.

Normalisation, logarithmisation, regression and scaling were then applied as the next pre-processing steps. The expression matrix was total-count normalised to 10,000 reads per cell in order to make counts comparable between cells, and a natural logarithm was applied to the normalized counts. The effects of total counts per cell and percentage of mitochondrial genes were regressed out to remove unwanted sources of variation. A filter for genes with maximum standard deviation of 10 was applied, and the counts for each gene were scaled to unit variance. In effect, this scaling ‘standardises’ the contribution of each gene to a cell’s gene expression profile and prevents genes with constant levels of high expression from having too large an influence on each cell’s overall profile.

It is also important to generate a lower-dimensional representation of the complete gene expression space. Here, principle component analysis (PCA) was applied to the whole dataset to generate a 50-component representation of the full gene expression space. For analyses that involve calculating gene expression distances between cells, this step is crucial. Because the full feature space of the scRNA-seq data consists of roughly 15,000 genes, cells in the raw data set suffer from the so-called “curse of dimensionality” (Bellman, 1961), a mathematical phenomenon through which distance metrics become ineffective in high-dimensional spaces, with all points being roughly equally distant from each other. The PCA reduces the data to a number of dimensions that allows the relative distances between cells to become distinguishable. Furthermore, the PCA discovers the dimensions in the original data set with the largest variance in gene expression, which are more likely to be biologically significant. The variance ratio of a principle component (PC) is the ratio between the variance for that

PC, and the total variance using all PCs. It indicates the proportion of the total variance that is explained by that PC. To check how much variance can be explained by the 50 PCs used in the PCA, variance ratios of the top 50 PCs were computed. Here, we calculated by adding top 50 PC variance ratios, that the 50-component representation of the gene space explained 6% of the total variance (Fig. 2. D).

Initial analysis revealed that *Malat1* was the top highly expressed gene, meaning that it yields the highest mean percentage of total counts across all cells. This is consistent with previous findings of *Malat1* detection, including RNA-seq of the mouse liver and brain cortex (Zhang *et al.*, 2012), scRNA-seq of human lung, spleen and oesophagus tissue (Madisson *et al.*, 2020) and DroNc-seq of hiPSC-derived cardiomyocytes (Selewa *et al.*, 2020).

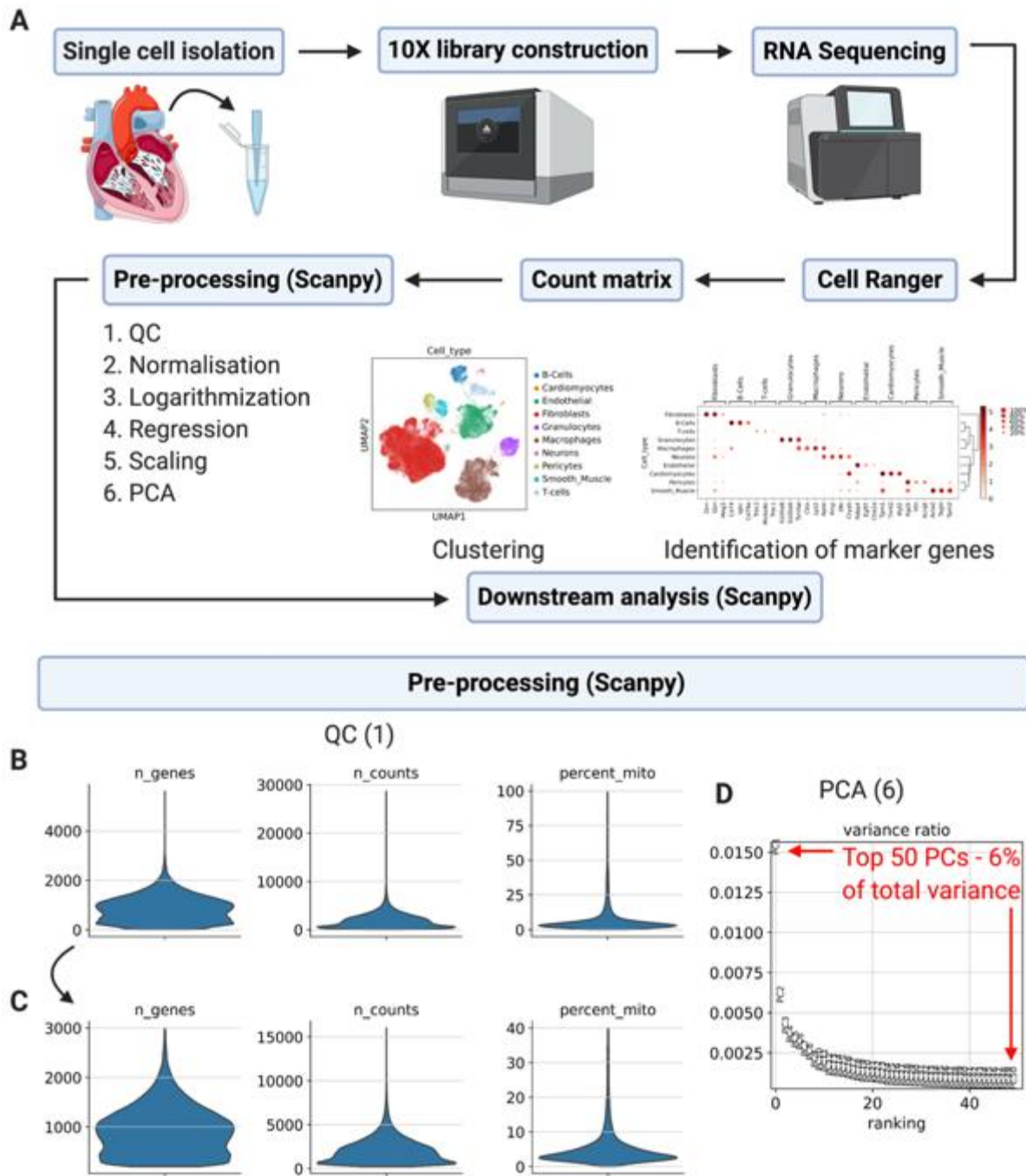


Figure 2. Pre-processing and analysis of raw data. A) Schematic diagram of steps that were taken from single cell isolation to downstream analysis. Pre-processing involved QC (quality control), normalisation, logarithmization, regression, scaling and PCA (principle component analysis). Pre-processing and downstream analysis were performed with *Scanpy*. Downstream analysis plots are examples of the results (clustering plot from Fig. 3. E; identification of marker genes plot from Fig. 3. H). For QC, filtering was applied: $200 < n_genes < 3000$, $percent_mito < 40\%$, $n_cells > 50$ (n_genes : number of genes expressed per cell; $percent_mito$: percentage of mitochondrial genes per cell; n_cells : number of cells that express a gene). Violin plots showing the distribution of cells with n_genes , n_counts and $percent_mito$ **B)** before filtering and **C)** after filtering (n_counts : number of total counts per cell). **D)** PCA variance ratios of the top 50 PCs (principle components). The top 50 PCs explain 6% of the total variance (calculated by adding top 50 PC variance ratios, multiplied by 100). Numbers next to QC and PCA correspond to their pre-processing step number in **A)**. (Figure created with BioRender.com).

Integration of batches

To integrate the 3 batches of data including different timepoint post-MI (Fig. 3. A), we applied correction of batch effects to the whole dataset using *BBKNN*. The integration of the batches and timepoints were then visualised by a graph-based dimensionality reduction technique, Uniform Manifold Approximation Projection (UMAP). Importantly, UMAP preserves local similarity between cells but does not give a true representation of global similarity like PCA. In other words, it clusters similar cells together. UMAP provided an effective 2-dimensional representation of the data, minimal batch effects and proper mixing of timepoints, as indicated by a representation of all batches (Fig. 3. B) and all timepoints (Fig. 3. C) in the various clusters of similar cells.

The overall cardiac cellular landscape

Annotation of the clusters from the global objects (kindly provided by provided by Dr. Almeida Miranda) allowed the identification of 10 major cell types in the complete dataset based on the expression of well-known marker genes: fibroblasts (*Dcn*); endothelial (*Fabp4*, *Cdh5* and *Pecam1*), pericytes (*Vtn*); smooth-muscle (*Myh11*); macrophages (*Cd68*); cardiomyocytes (*Tnnt2*); B-cells (*Cd79a*); granulocytes (*S1009a*); T-cells (*Trbc2*); neurons (*Kcna1*). Fibroblasts were the largest cluster, with the highest number of cells (42,181), representing overall 56% of the total cells captured. Representation of the main cell types on a UMAP showed grouping of fibroblast, pericyte, smooth-muscle, T-cell, B-cell and endothelial clusters, whereas the neuron, cardiomyocyte, macrophage and granulocyte clusters were separated from this group (Fig. 3. E).

To study possible changes in cell distribution in healthy hearts and after coronary artery ligation, the distribution of each cell type was calculated as the proportion of total cells present at each timepoint (Fig. 3. D). Fibroblasts were the most abundant cell type across all timepoints except D03 post-MI, which comprised mostly macrophages. The lack of fibroblasts at D03 could be due to the fact that only the Harvey batch contributed towards the D03 timepoint, and its fibroblasts were not enriched. Hierarchical clustering was used to compute a population dendrogram, which revealed that B-cells and T-cells are the most similar and fibroblasts are the least similar to any other cell type (Fig. 3. F). The hierarchical clustering takes the mean of each cluster (or in this case, cell type) and treats this as a single cell. Therefore, this method does not consider the variance of gene expression values within the clusters. It constructs a dendrogram based on the Pearson's correlation coefficients between the cluster means. This reveals the relationship between clusters in a population across all 50 top PCs, providing an indication of global similarity across the expression values of all genes.

Comparison of methods to define differentially expressed genes

Before proceeding to sub-clustering of fibroblasts to define the heterogeneity of identified cell sub-types/states and make functional inferences, it was crucial to define the best method for calculating of differentially expressed genes (DEGs). 3 statistical methods are commonly used: Wilcoxon Rank-Sum test (Wilcoxon), logistic regression (logreg), and t-test. Here, only the results from the logreg and Wilcoxon methods were investigated in detail in downstream analyses, to avoid applying the t-test in any situations where its parametric assumptions (namely, that the logarithmized count data are normally distributed) were violated. The top 100 DEGs were identified within

the fibroblasts out of the complete dataset using logreg, Wilcoxon and t-test methods, and the number of overlapping or unique DEGs were calculated and represented in a Venn diagram (Fig. 3. G). The logreg method returned mostly unique DEGs compared to those obtained with the Wilcoxon method. This is to be expected as the logistic regression uses a multivariate model that simultaneously considers the expression of all genes at once, controlling for the effects of correlated genes. On the other hand, the Wilcoxon test is a univariate method that does not consider the correlation between genes, and instead separately tests every gene for association with the groups of interest.

For identification of DEGs analysed and visualized in downstream sections, we chose to apply the logistic regression method. In addition to the fact that the method controls for the effects of other genes in assessing a gene's statistical significance, qualitative inspection revealed that the genes identified by logreg appeared to be more unique to the manually annotated cell populations than those identified by Wilcoxon. All gene expression dotplots in later figures were therefore produced using the results of logreg. As a complementary approach, the Wilcoxon method was used to identify DEGs that were subsequently analysed by ToppGene in order to make inferences regarding subpopulation functions. Because ToppGene's functional annotations rely on the co-expression of multiple genes to identify enriched pathways and processes, the Wilcoxon was deemed a more effective test than logistic regression for the fact that it does not "dampen" the effect size of genes that correlate highly with other genes, as the logreg does. All in all, logistic regression was assumed to provide a more effective way to identify individual marker genes that define cell subpopulations, while for identifying enriched biological functions in different cell subpopulations, the Wilcoxon was deemed more appropriate.

Cardiac cell-type-specific markers

In order to identify unique cell-type markers, the top 3 DEGs were calculated across all the cell types in the complete dataset using the logreg method and visualised with a dotplot (Fig. 3. H). Genes tested here for differential expression were filtered to those with a minimum log fold change of 2. Notably, *Dcn* was the top gene expressed in fibroblasts, with high specificity to this cell type. *Dcn*, which encodes the Decorin protein, regulates various cellular functions such as angiogenesis, inflammation and ECM organisation (Lai *et al.*, 2017). More specifically, *Dcn* null mice show increased scar size, right ventricular hypertrophy (abnormal enlargement of the heart muscle) and left ventricular dilation than wild types, indicating that it plays a key role in fibrosis following MI (Weis *et al.*, 2005).

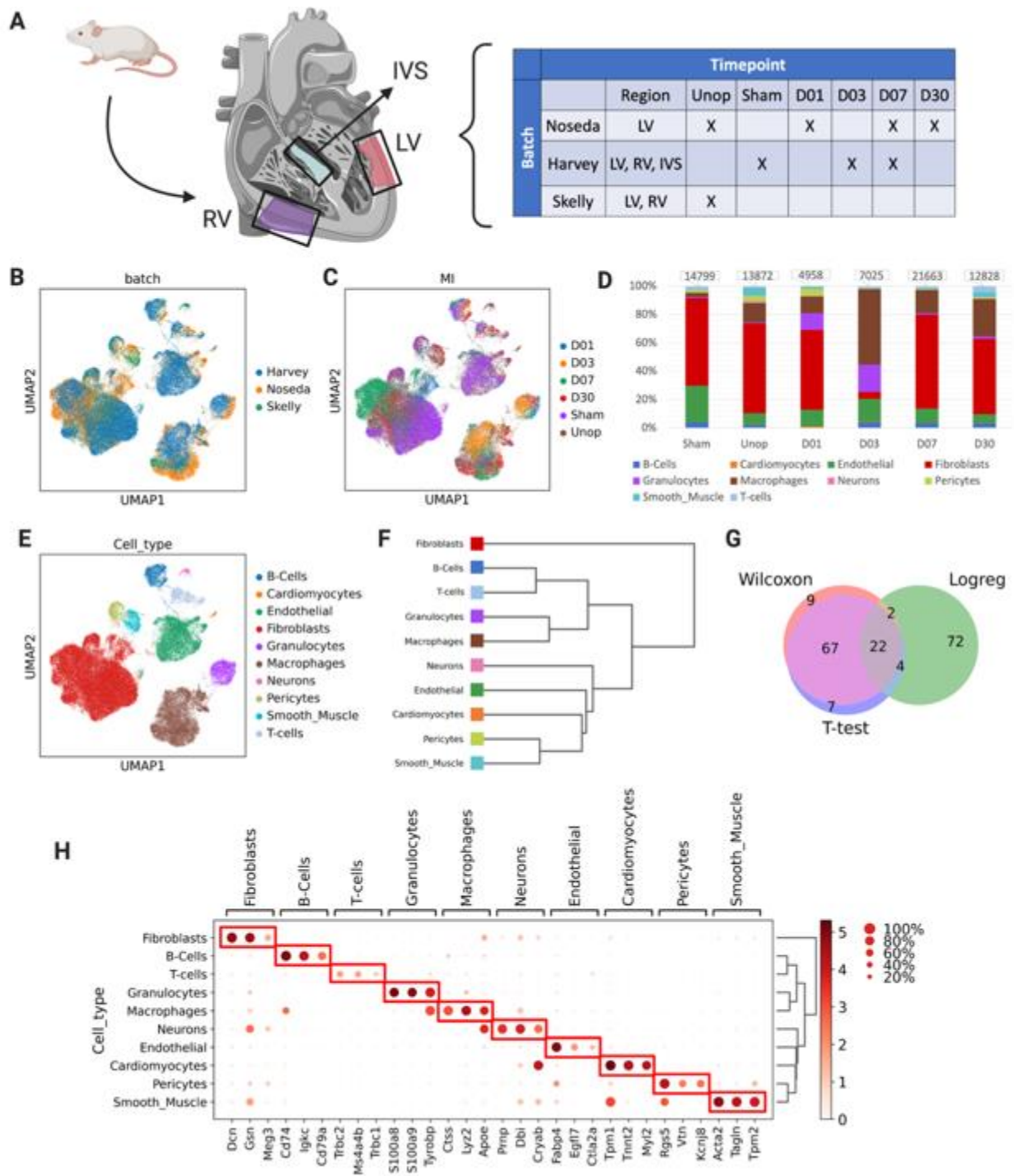


Figure 3. The global objects representing the overall cellular landscape of the hearts analysed. **A)** Schematic diagram showing the lab of origin, ventricular region from which the cells were isolated and timepoints, for each batch. (LV, left ventricle; RV, right ventricle; IVS, interventricular septum). UMAP (uniform manifold approximation projection) representing the global objects including 75145 single cells and illustrating covariates for **B)** batches and **C)** timepoints within the whole dataset. **D)** Stacked barplot showing the distribution of cell types across timepoints. Numbers above the bars show the total number of cells for that timepoint. **E)** UMAP of main cell types identified in the whole dataset, after Leiden clustering. **F)** Population dendrogram of cell types, based on hierarchical clustering. **G)** Venn diagram of overlapping DEGs (differentially expressed genes) identified in fibroblasts using 3 different methods: Wilcoxon (Wilcoxon rank-sum), logreg (logistic regression) and T-test. **H)** Dotplot showing the top 3 upregulated genes for each cell type with dendrogram (method=logreg, filtered to minimum log fold change=2). Colour bar represents mean change in expression of the gene across all cells in the cell type. Dot size and percentage represents the percentage of cells that express the gene within the cell type. (Figure created with BioRender.com).

Clustering optimisation

When clustering data, there are multiple important considerations in choosing the “optimal” clustering. The ideal clustering produces groups of data points that are well-separated from other groups and tightly connected within groups. In the context of scRNA-seq data, we hope to identify homogenous populations of cells that are easily distinguishable from other populations (based on gene expression). However, it should be noted that it is impossible to identify a completely “optimal” clustering for a given problem. This is due in part to experimental noise, like reverse-transcription dropouts, which produce data that are fundamentally incorrect. However, even in a perfect scRNA-seq data set, no “optimal” group structure exists objectively because biology itself is hierarchically organised into populations and subpopulations of cells at different levels of detail. It is therefore helpful to view clustering algorithms not as a way to uncover some globally optimum population structure, but rather as a way to identify an optimal group structure, given a data set and a resolution (level of detail) at which the data should be viewed. Nevertheless, when exploring cell type heterogeneity, it is important to fine-tune the resolution of the clustering to identify group structures at levels of detail that are biologically meaningful. To fine-tune the clustering resolutions, one can use qualitative approaches that involve mapping differentially expressed genes in clusters to published literature or calculate unbiased quantitative metrics to identify that capture the quality of the clustering at that resolution.

In order to compare multiple possible resolutions of fibroblast sub-clustering, the Leiden clustering algorithm was applied to cardiac fibroblasts within the complete dataset, and the resolution parameter was changed each time. Unbiased quantitative metrics, mentioned previously, were used to assess the quality of the clustering at 3

example resolutions. The fibroblast objects were extracted from the dataset (Fig. 4. A) and Leiden was run with resolutions 0.2, 0.4 and 1.1, producing 3, 7 and 17 clusters respectively (Fig. 4. B). Increasing the resolution resulted in a larger number of clusters as expected. To assess the quality of the clustering, multiple methods were used at each resolution, such as generation of Pearson correlation matrix plots, analysis of Receiver Operator Characteristic (ROC) curves from the Single Cell Clustering Assessment Framework (SCCAF) (Fig. 4. C-E).

Correlation matrix plots were made for each of the 3 resolutions after clustering and were based on hierarchical clustering, as with the dendrogram. Each value on the plots denotes the Pearson's correlation coefficient, and therefore the similarity, between the pair of clusters. This revealed that the highest correlation between any 2 clusters was reached at resolution 1.1 (Fig. 4. C). Particularly, cluster 0 and cluster 11 at resolution 1.1 had a similarity value of 0.69, indicating that these clusters could comprise of cells of the same sub-type. As clustering of fibroblasts was performed in order to characterise the heterogeneity of the sub-types of cells among fibroblasts, we decided that resolution 1.1 did not separate the subclusters well.

ROC curves were made with the SCCAF package to further assess the performance of the Leiden clustering at each resolution (Fig. 4. D). SCCAF trains a logistic regression model (classifier) to distinguish cells within each cluster from cells outside of each cluster, providing an indication of how well-separated each cluster of cells is from the other cells. The axes of figure 3 show the False Positive Rate (FPR) plotted against True Positive Rate (TPR) of the classifier for each cluster, which forms a curve that informs of the classifier's performance for each cluster (and thus of the quality of each individual cluster). The larger the Area Under Curve (AUC) for a specific cluster, the lower the ratio of false positives to true positives and therefore the more

accurate the model is at classifying this cluster. Here, they indicated that the clustering at resolution 0.2 allowed for the most accurate classification of its clusters overall, with a minimum and maximum AUC of 0.930 and 0.998, respectively. The cluster with the highest AUC from each resolution was identified: cluster 2 at resolution 0.2 (AUC=0.998), cluster 6 at resolution 0.4 (AUC=0.996) and cluster 16 at resolution 1.1 (AUC=0.996). These can be thought of as the most distinguishable clusters at each resolution.

To further assess the quality of the clustering at each resolution, Calinski-Harabasz (CH) and Silhouette scores were also calculated and plotted for the resolution range 0.2-2.1 (Fig. 4. E, F). Calinski-Harabasz scores represent the ratio of the sum of between-clusters dispersion and of within-cluster dispersion for all clusters (where dispersion is defined as the sum of distances squared). However, Silhouette scores are calculated by comparing within-cluster cohesion to the distance to the nearest cluster (using the mean distance between a cell and all other cells in the same cluster, and the mean distance between the cell and all cells in the next nearest cluster). Both methods result in higher scores when clusters are dense and well-separated. In this analysis, CH and Silhouette scores decreased as resolution was increased, as expected. Both scores had 2 peaks at resolutions 0.4 and 1.1, indicating that either the 7- or 17-cluster solution might produce the most well-separated subclusters.

A final resolution for fibroblast sub-clustering of 0.4 was reached by taking these metrics into account; its correlation matrix plot had no coefficients above 0.5, its maximum and minimum AUCs were above 0.9 and both CH and Silhouette scores peaked at this resolution. In terms of gene expression, this resolution produced subclusters that showed the most subcluster-specific markers compared to the other resolutions (described in the following section).

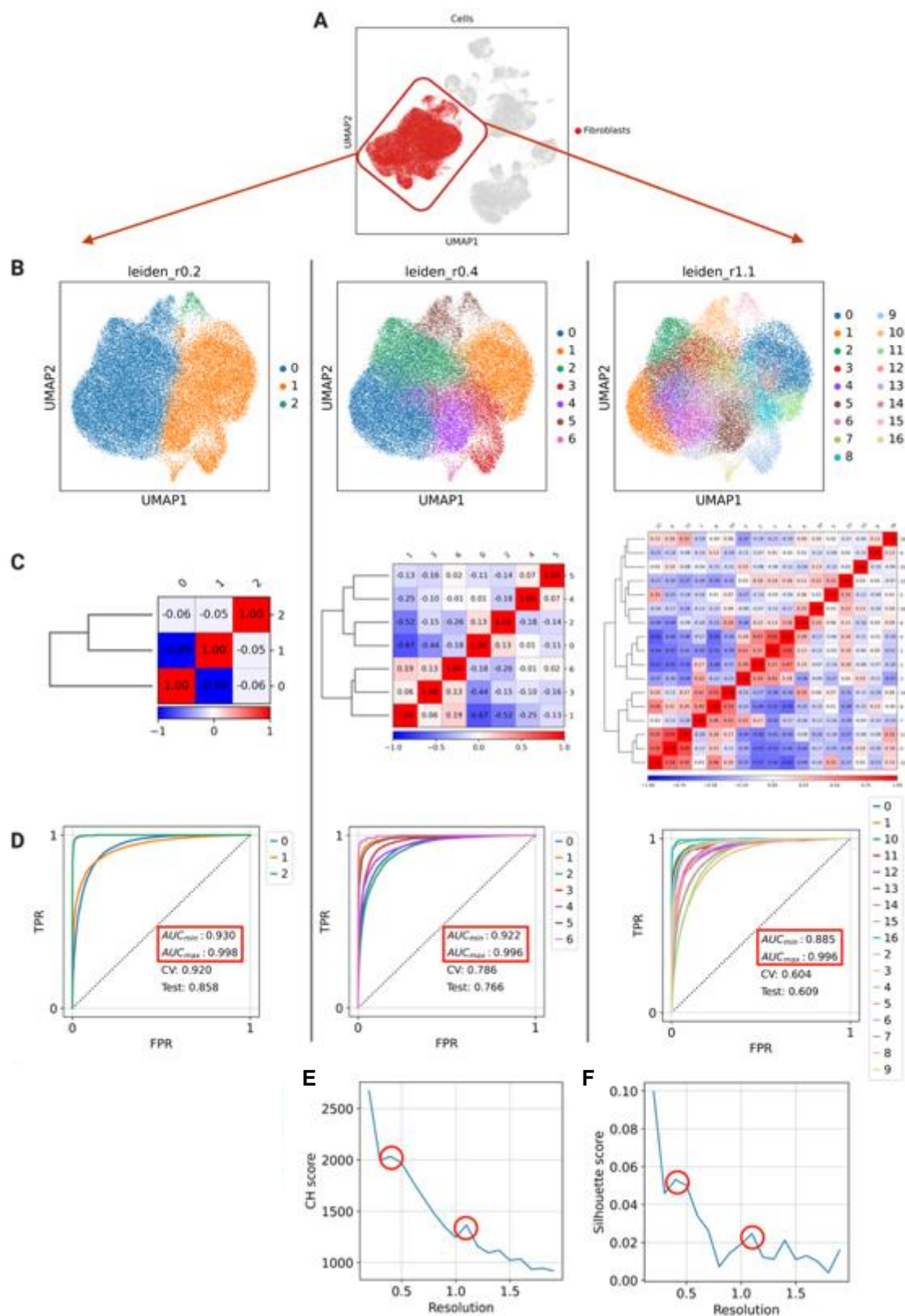


Figure 4. Clustering optimisation. **A**) UMAP (uniform manifold projection approximation) of whole dataset with fibroblasts coloured. **B**) UMAPs showing differences in subcluster number after Leiden clustering, between resolutions 0.2, 0.4 and 1.1. **C**) Pearson's correlation matrix plots for each resolution. Values represent similarity values between subclusters. **D**) ROC (receiver operator characteristic) curves made using the SCCAF package (single-cell clustering assessment framework), at each resolution (AUC, area under curve; CV, cross-validation accuracy; Test, test accuracy). **E**) CH (Calinski-Harabasz) scores and **F**) Silhouette scores for resolutions between 0.2 and 1.9 (0.1 increments). Peaks showing the best clustering resolutions are circled red. (Figure created with BioRender.com).

Heterogeneity of cardiac stromal fibroblasts

To characterise the heterogeneity of the cardiac stromal fibroblasts from both healthy and post-MI hearts, a final resolution of 0.4 was chosen for Leiden sub-clustering, which produced 7 subclusters (Fig. 5. A). Hierarchical clustering, using Pearson correlation coefficients, showed 2 pairs of clusters of equal similarity: F1 and F3; F4 and F7, and these pairs were the most similar compared to the other fibroblast subclusters (Fig. 5. B). This suggested that F1 and F3 express similar genes and therefore have similar functions in the heart, as with the case of F4 and F7.

In order to identify the most unique markers for these fibroblast subclusters, the top 3 DEGs across the subclusters were obtained with the logreg method (Fig. 5. C). Here, no filtering was applied. Subcluster F2 showed high expression of *Thbs4* encoding Thrombospondin 4, which plays a critical role in regulating ECM production and organisation (Cingolani *et al.*, 2011). Notably, *Wif1* was the top gene expressed in F4. *Wif1* encodes a secreted canonical WNT pathway inhibitor which regulates the inflammatory response after cardiac injury (Meyer *et al.*, 2017), hence F4 cells may contribute to this response.

Plotting expression of the top DEG (identified with the logreg method) for each subcluster on a UMAP indicated that whilst these markers are uniquely expressed in their corresponding subclusters, they are not expressed in all of the cells within subclusters (Fig. 5. D). In particular, by this method, *Wif1* was only expressed in the bottom half of F4 cells, and *Stmn1* was only expressed in the right half of cells of F6. This suggests that there might be further heterogeneity within these fibroblast subclusters.

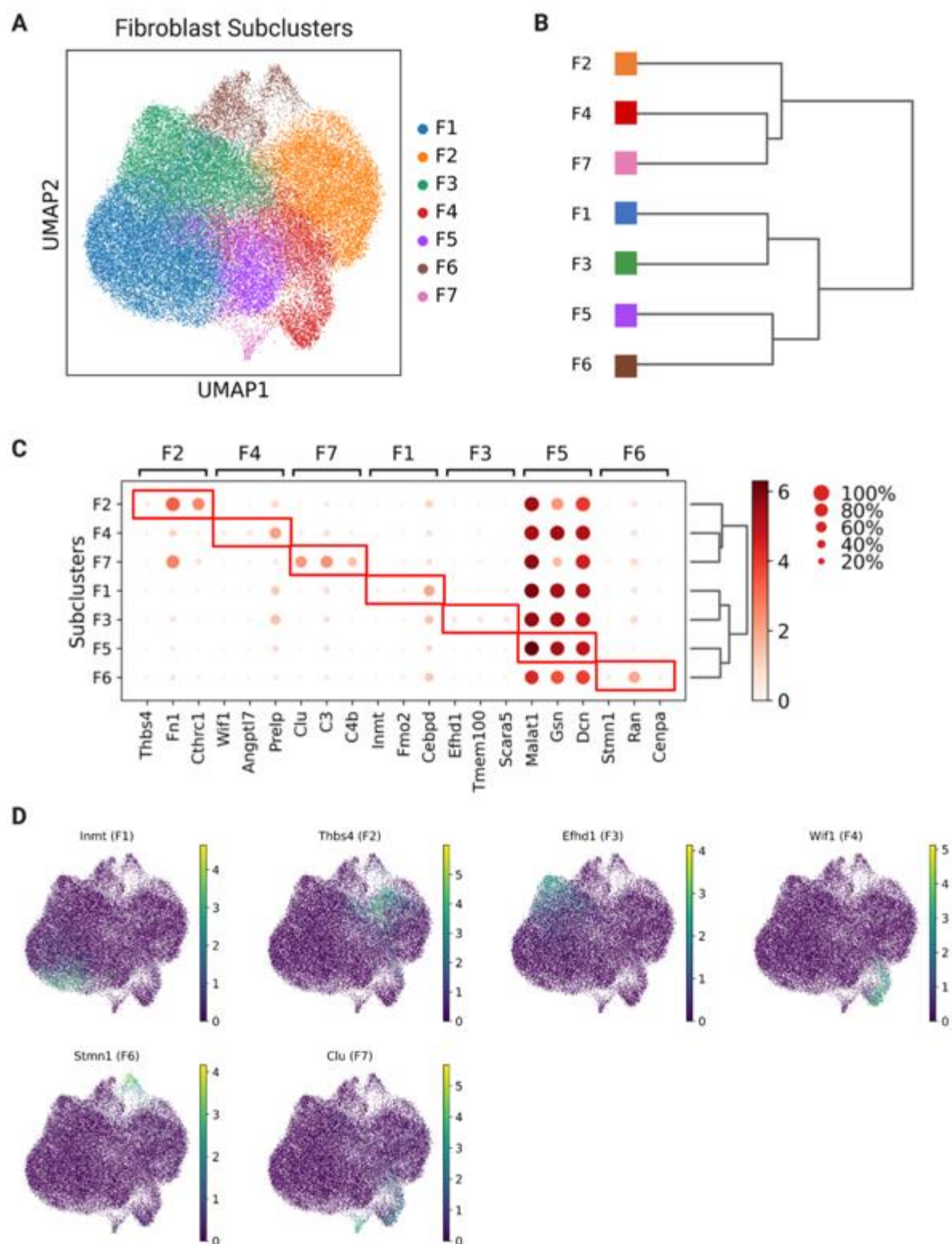


Figure 5. Heterogeneity of cardiac fibroblast cells. **A)** UMAP (uniform manifold approximation projection) of fibroblast subclusters F1-F7, after clustering with Leiden (resolution=0.4). **B)** Population dendrogram of fibroblast subclusters, based on hierarchical clustering. **C)** Dotplot of the top 3 upregulated genes in the subclusters with dendrogram (method=logreg, logistic regression). Colour bar represents mean change in expression (log fold change) of the gene across all cells in that subcluster, compared to other subclusters. Dot size and percentage represents the percentage of cells that express the gene in that subcluster. **D)** Gene expression UMAPs of subcluster-specific markers. Colour bar represents log fold change expression. (Figure created with BioRender.com).

Annotation of fibroblast subclusters and inferences in functions

Previous analysis from the Nosedá team, performed using Smart-seq2 following index-sorting, identified a cluster of SP cells enriching for *Dkk3*, *Scara3* and *1500015O10Rik* (unpublished). This cluster is part of a larger subpopulation of cMSCs expressing three flow cytometry features in combinatory fashion (Nosedá *et al.*, 2015). This Sca1⁺/Pdgfra⁺/SP subpopulation exhibits properties such as high clonogenicity and highly uniform cardiogenic gene expression (*Gata4/6*, *Hand2*, *Tbx5/20*) (Fig. 6. A).

Visualisation of mean expression of these previously described markers on a dotplot (Fig. 6. B) and DEG calculation with the Wilcoxon method showed highest expression of *Dkk3*, *Scara3* and *1500015O10Rik* in F4. Expression of *Wif1* relative to these markers can also be seen; its expression is relatively low in all subclusters but specific to a small proportion of cells in F4.

For further identification of the global specificity and heterogeneity of *Wif1* expression within the F4 subcluster, F4 was first extracted from the fibroblasts (highlighted globally Fig. 6. C; highlighted among fibroblast subclusters Fig. 6. D). As mentioned previously, *Wif1* is highly expressed in subcluster F4 compared to the other subclusters within fibroblasts but is also exclusively expressed in this subcluster when plotted globally (Fig. 6. E). Within F4, *Wif1* is expressed in only a fraction of F4 cells (expression highlighted within fibroblasts Fig. 6. F; highlighted within the F4 subcluster Fig. 6. G). To identify differences in *Wif1* expression among batches, *Wif1* expression was plotted against batches within F4 on a UMAP (Fig. 6. H), which revealed that *Wif1* is represented by all 3 batches. Further comparison of the distribution of *Wif1* expression in unoperated fibroblasts (expression>0, batches Nosedá and Skelly together) showed that *Wif1* is collectively expressed in 3% of these cells. However,

the Skelly batch had a three-fold proportion of cells that express *Wif1* (6%), compared to the Nosedá batch (2%).

Additional relevant markers were also visualized (Fig. 6.B). *Pdgfra* showed overall low expression across all subclusters and *Ly6a* (Sca1) showed high mean expression in more than 70% of cells in the F3 subcluster. These markers were also identified using the Wilcoxon method (q-value<1E-5, log fold change>1). Using this method, *Pdgfra* was differentially expressed in F1 and *Ly6a* (Sca1) was differentially expressed in F3. This indicates that F3 might correspond to the *Pdgfra*-GFP⁺ fibroblast population F-SH previously identified (Farbehi *et al.*, 2019). Cardiogenic markers *Tbx5*, *Gata4* and *Hand2* showed low expression across all clusters and were not differentially expressed. However, *Gata6* was differentially expressed in F7 (q-value<1E-5, log fold change>1). *Cdh5* and *Kdr*, previously described in the Sca1⁺ and non-SP population, showed low expression and were not differentially expressed across the subclusters.

In order to make inferences in fibroblast subcluster functions, ToppGene was run on the subcluster DEGs and the top GO terms of biological processes were visualised in a heatmap (script provided by Dr. Michael Lee, Fig. 6. I). GO terms associated with F1 included negative regulation of cell population proliferation, although the highest number of DEGs matched with enzyme linked protein receptor signalling. F2 had 7 biological processes, with over 60 DEGs associated with ECM and structure organization, skeletal system development, supramolecular fiber organisation, circulatory system development and animal organ morphogenesis. The former 3 biological processes were also associated with F4 and the latter 2 were associated with F7, with fewer matched DEGs. F3 was associated with cell morphogenesis, cell morphogenesis regulation, regeneration and cellular component morphogenesis. Connective tissue development and cartilage development were exclusively

associated with F4. The top 5 processes for F5 were oxidative phosphorylation, ATP synthesis coupled protein transport, energy coupled proton transport (down electrochemical gradient), ATP biosynthetic process and purine ribonucleoside triphosphate biosynthetic process. Interestingly, DEGs in F6 were associated with regulation of hematopoietic stem cell differentiation. Vasculature development was uniquely associated with F7.

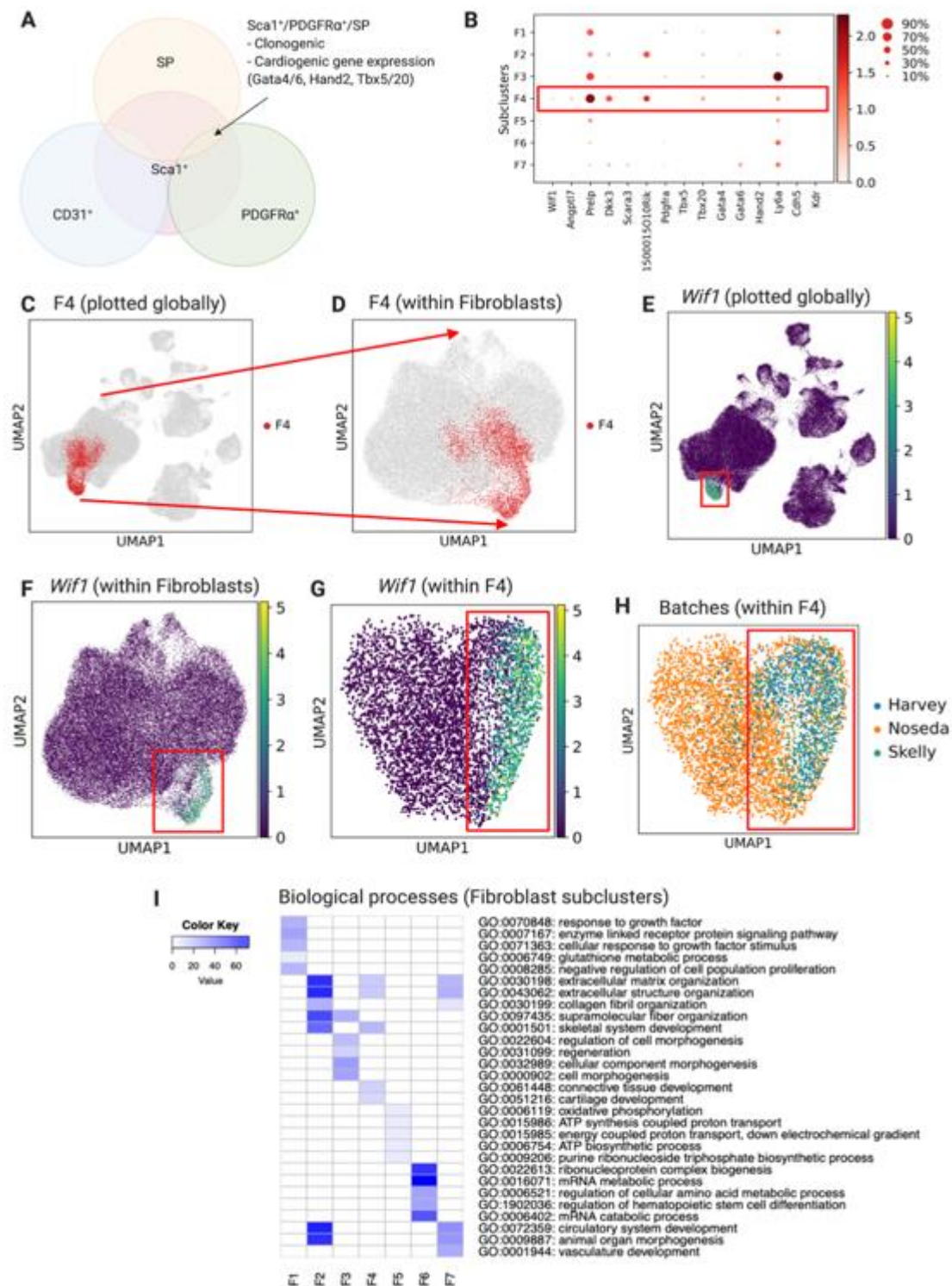


Figure 6. Annotation of fibroblast subclusters and inferences in functions. **A)** Venn diagram of previously described subpopulations of cMSCs (cardiac mesenchymal stromal cells) by the Nosedá team (Nosedá *et al.*, 2015). SP (side population). **B)** Dotplot showing expression of previously described markers in the fibroblast subclusters. Colour bar represents mean expression of the gene across all cells in that subcluster. Dot size and percentage represents the percentage of cells that express the gene in that subcluster. UMAP (uniform manifold approximation projection) of fibroblast subcluster F4 (coloured red) plotted **C)** globally and **D)** within fibroblasts. UMAP showing *Wif1* expression **E)** globally, **F)** within fibroblasts and **G)** within F4. Colour bar represents mean log fold change of the gene in each cell/dot. **I)** Heatmap of the top biological processes for fibroblast subclusters (made using script written by Dr. Michael Lee). Value/colour in the colour key shows the number of genes in the DEG (differentially expressed genes) list that have the GO (gene ontology) term associated with it. (Figure created with BioRender.com).

Phenotypic and functional changes post-MI

To understand the changes occurring to the fibroblasts compartment post-MI, UMAPs were generated to visualize the distribution of fibroblasts from specific time points (Fig 7. A - F) representation supported the here were very few fibroblasts in D03 post-MI hearts (Fig. 7. D), likely due to the lack of enrichment for fibroblasts at this timepoint, as mentioned previously. Fibroblasts present in the unoperated hearts (Fig. 7. A) were most similar to cells in sham-operated hearts (Fig. 7. B) as shown by an overlap of the two sources and supported by computation of a population dendrogram based on hierarchical clustering (Fig. 7. G).

To study possible changes in fibroblast sub-type distribution in healthy hearts versus after coronary artery ligation, the distribution of each fibroblast subcluster was calculated as the proportion of total fibroblast cells present at each timepoint (Fig. 7. H). Notably, there was a dramatic increase in the number of F2 cells at D07 post-MI.

To investigate F2 gene expression changes across timepoints, the top 3 DEGs for the F2 subcluster across all timepoints were identified (logreg method) and visualised with a dotplot (Fig. 7. I). The top DEGs with the logreg method at D07 were not unique to this timepoint, but those identified using the Wilcoxon method (not shown) included collagen genes, important for ECM organisation. Using the latter method, this subcluster also expressed *Postn* and *Acta2*, indicating that these cells could be myofibroblasts which have been previously shown to contribute towards collagen deposition.

To identify potential F2 functions post-MI, ToppGene was run on the timepoint DEGs and the top GO terms of biological processes were visualized on a heatmap (Fig. 7. J). Interestingly, processes such as ECM organization and structure

organization were upregulated at D07, which agrees with the DEGs found with Wilcoxon method and the annotation for this subcluster as myofibroblasts.

Changes in subcluster F4 post-MI

In order to investigate the distribution of the SP cells across the 6 timepoints, cells present at each timepoint were plotted together on a UMAP (Fig. 8. A). Cells present at all timepoints correlated with the group of cells from all batches (please refer back to Fig. 6. H). A large number of cells were present at D30 only. Expression of previously described markers from the $Sca1^+/Pdgfra^+$ /SP subpopulation (please refer back to Fig. 6. B) were visualised again across the timepoints, within SP cells (Fig. 8. B).

In order to identify SP gene expression changes post-MI, the top 3 DEGs were calculated across the timepoints using logreg (Fig. 8. C). DEGs were filtered to a minimum log fold change of 0.5.

DEGs within F4 cells across timepoints were calculated again using the Wilcoxon method ($q\text{-value} < 1E-5$, $\log\text{ fold change} > 1$). To identify potential F4 functions post-MI, ToppGene was run on the timepoint DEGs and enriched pathways were visualized on a heatmap (Fig. 8. D). The value shown in the color key represents the number of terms in the DEG list that match the pathway divided by the total number of terms associated with that pathway. No genes were differentially expressed at Day 3, so no categories/pathway could be calculated. A pathway identified for D07, C-MYB transcription factor network, is suggestive of a switch/transition of function at this timepoint.

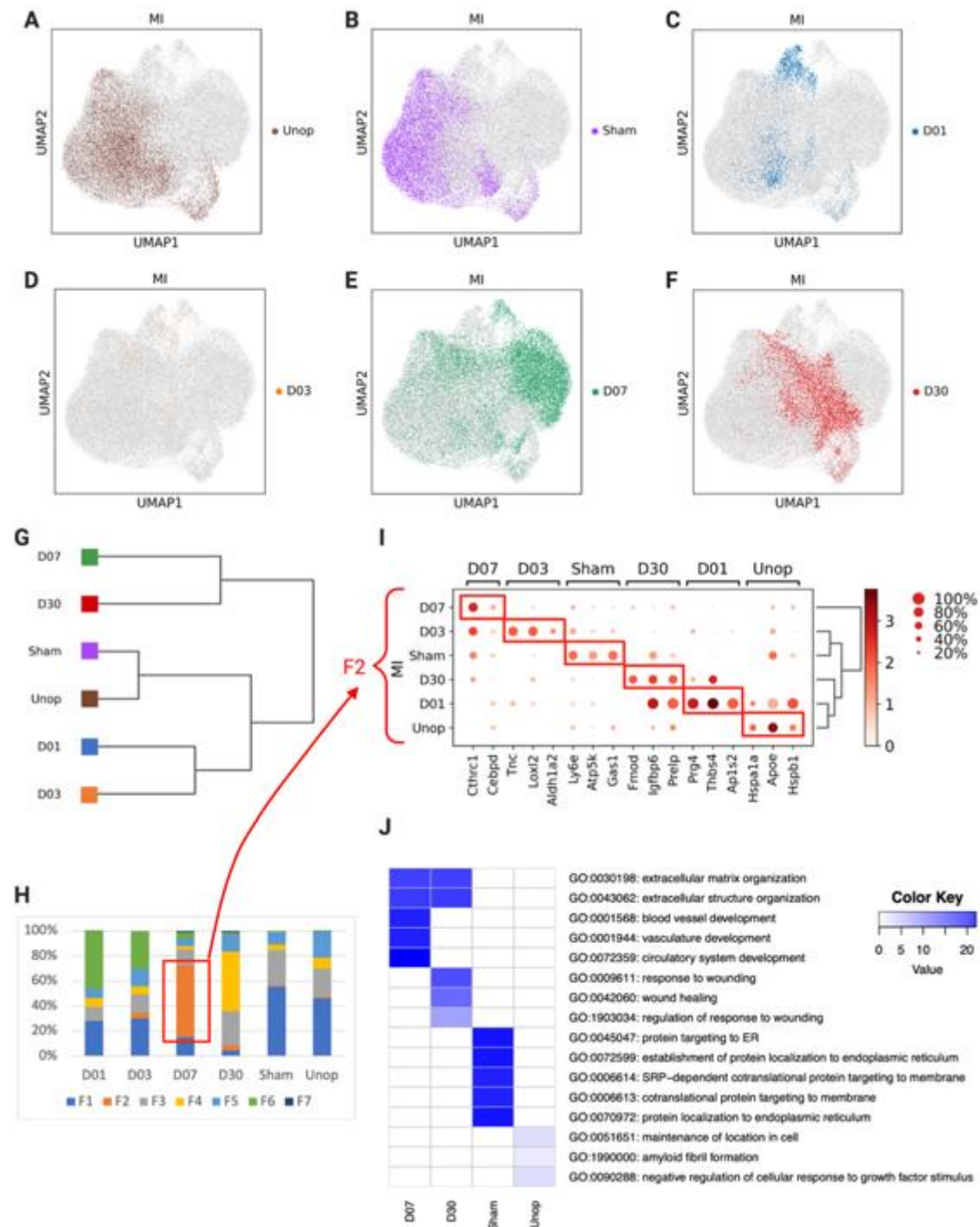


Figure 7. Phenotypic and functional changes in fibroblasts post-MI (myocardial infarction). UMAP highlighting the cells from individual sources **A)** Unop (unoperated), **B)** Sham (sham-operated), **C)** D01, **D)** D03, **E)** D07 and **F)** D30. **G)** Population dendrogram of timepoints, based on hierarchical clustering. **H)** Stacked barplot showing the distribution of fibroblast subclusters across timepoints (normalized to total fibroblast cells per timepoint) with F2 highlighted (red box). **I)** Dotplot showing the top 3 upregulated genes within F2, across timepoints (method=logreg, logistic regression; filtered to a minimum log fold change=0.5). Colour bar represents mean change in expression (log fold change) of the gene across all cells in that timepoint. Dot size and percentage represents the percentage of cells that express the gene in that timepoint. **J)** Heatmap of the top biological processes within fibroblasts across timepoints (made using script written by Dr. Michael Lee). Value/color in the color key shows the number of genes in the DEG (differentially expressed genes) list that have the GO (gene ontology) term associated with it. (Figure created with BioRender.com).

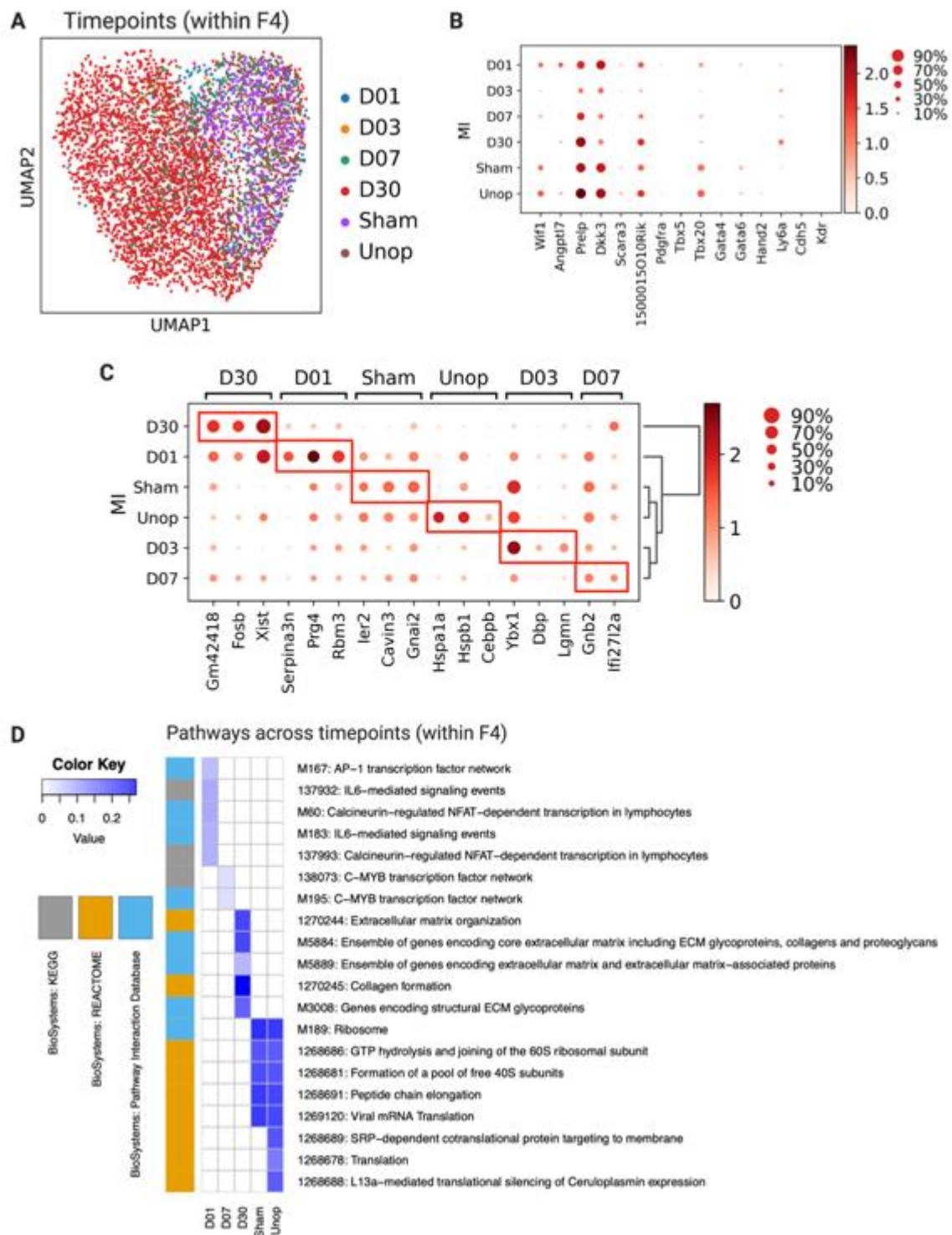


Figure 8. Phenotypic and functional changes in F4 post-MI (myocardial infarction). **A)** UMAP (uniform manifold approximation projection) of the distribution of timepoints within F4. **B)** Dotplot showing expression of previously described markers, within F4 and across timepoints. Colour bar represents mean expression of the gene across all cells in that timepoint. **C)** Dotplot showing the top 3 upregulated genes within F4, across timepoints (method=logreg, logistic regression; filtered to a minimum log fold change=0.5). Colour bar represents mean change in expression (log fold change) of the gene across all cells in that timepoint, compared to other timepoints. Dot size and percentage represents the percentage of cells that express the gene in that timepoint. **D)** Heatmap of the top pathways within F4 across timepoints (made using script written by Dr. Michael Lee). Value/colour in the colour key shows the number of terms in the DEG (differentially expressed genes) list that match the pathway, divided by the total number of terms associated with that pathway. No significant pathways were found for D03. (Figure created with BioRender.com).

Intercellular signalling between fibroblasts and macrophages

The Nosedá team has demonstrated cardiac protection in mice by intramyocardial injection of cloned SP cells following MI and identified their functions underpinning protection, including modulation of the fate of macrophages (unpublished data, obtained using same cMSC-conditioned media as in Constantinou *et al.*, 2020) which have a crucial dynamic role in response to injury. In short, after MI macrophages initially show a pro-inflammatory phenotype, or M1 state, and then undergo polarisation to their M2 reparative phenotype (Ma, Mouton & Lindsey, 2017).

To explore intercellular signalling between fibroblasts and macrophages computationally, macrophages from the whole dataset were first sub-clustered using the Leiden algorithm. Then, potential interactions between fibroblasts and macrophages were predicted based on the expression of interacting ligand-receptor pairs in the respective cell types, using *CellPhoneDB*. Further, changes in expression of receptors and ligands of interest across timepoints post-MI were visualised via dotplots.

Macrophages were sub-clustered into 12 subclusters using Leiden resolution 0.6 (Fig. 9. A). Computation of a population dendrogram based on hierarchical clustering indicated that out of the macrophage subclusters (M1-M12), M7 and M12 were the most similar (Fig. 9. B).

Global *CellPhoneDB* analysis of predicted signalling between cell types showed low mean expression of all signalling from macrophages to fibroblasts, but high expression of signalling fibroblasts to macrophages involving certain ligands and receptors. Specifically, macrophage-expressed receptor CD74 was predicted to bind with fibroblast-expressed ligands APP, COPA and MIF (Fig. 9. C). These 3 interactions were highly expressed in at least 60% of fibroblast and macrophage cells.

To investigate whether the top macrophage receptors are specific to this cell type and which subclusters they are highly expressed in, their expressions were visualised by dotplots. This showed that *P2ry6*, *Sirpa*, *Csf1r* and *C3ar1* were exclusively expressed in macrophages compared to other cell types (Fig. 9. D). On the other hand, *Cd74*, was more highly expressed in B-cells than macrophages. Within the macrophages, *Cd74* was highly expressed in subclusters M1, M7 and M12 (Fig. 9. E). Visualisation of *Cd74* expression across macrophage cells by UMAP representation supported this (Fig. 9. F). To explore whether *Cd74* expression within the macrophages differs between healthy and MI-induced hearts, its mean expression across the timepoints was visualised, which showed increasing expression from D01 through D03, D07 and D30, returning over time to levels seen in the Sham, to the Unop samples (Fig. 9. G). The expression patterns of *Cd74* alone tells us that *Cd74* expressing cells are temporally dynamic following MI.

MIF, the corresponding fibroblast ligand in one of the significant interacting receptor-ligand pairs involving CD74, has been demonstrated to bind to the soluble form of CD74 in response to heart failure (Soppert *et al.*, 2018). To explore whether *Mif* expression within the fibroblasts differs between healthy and MI-induced hearts, its mean expression across the timepoints was visualised, which revealed a pattern of expression that was opposite to that of *Cd74*, with little baseline expression and marked induction 1 day post-MI (Fig. 9. H). Thus, the expression patterns of *Mif* and *Cd74* across timepoints indicates that this receptor-ligand pair is highly dynamic across timepoints and in response to MI.

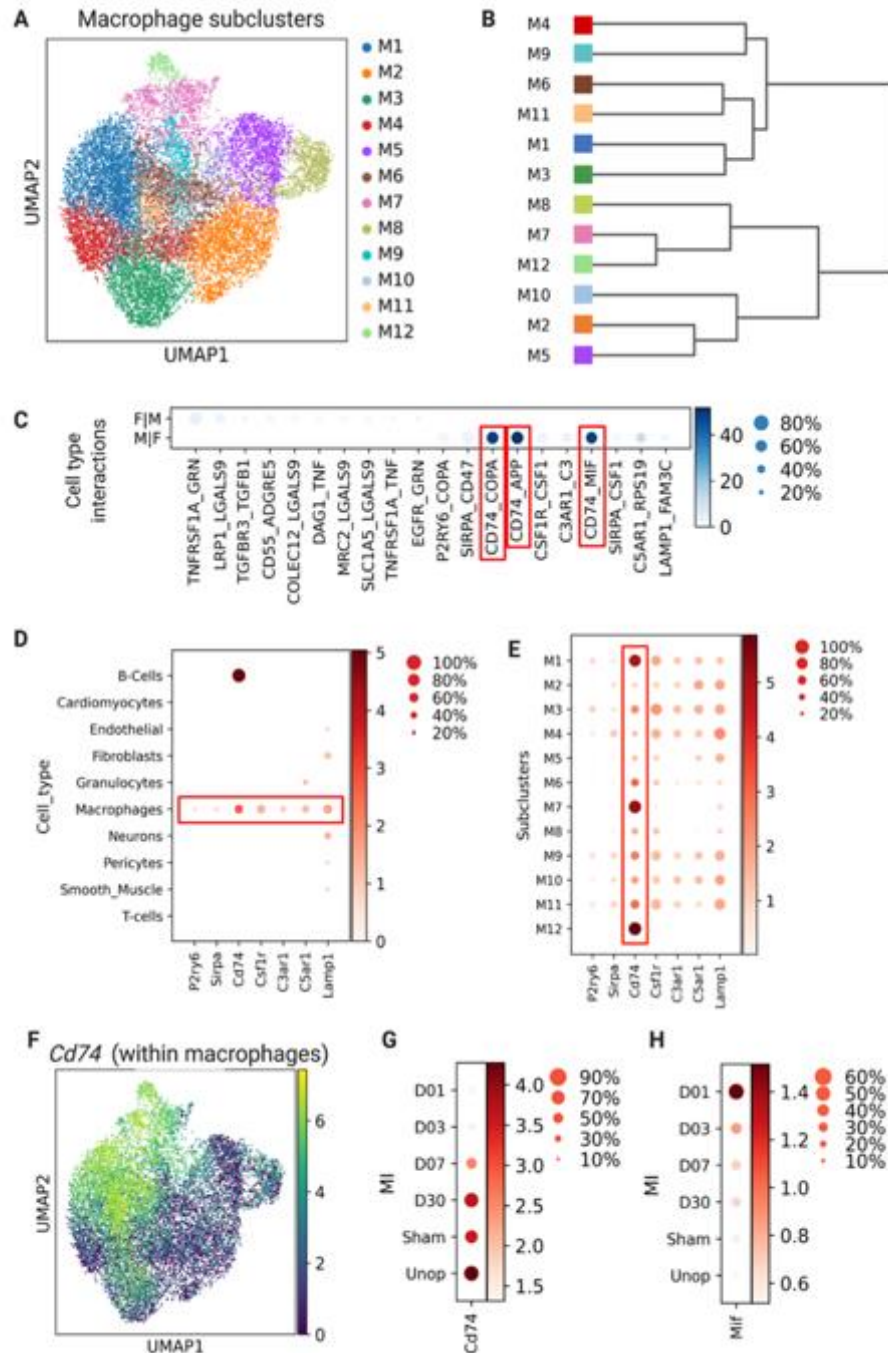


Figure 9. Intercellular signaling between fibroblasts and macrophages. **A)** UMAP (uniform manifold approximation projection) of macrophage subclusters M1-M12, after clustering with Leiden (resolution=0.6). **B)** Population dendrogram of macrophage subclusters, based on hierarchical clustering. **C)** Dotplot showing the top 20 interactions between fibroblasts and macrophages, predicted from global *CellPhoneDB* analysis. F|M represents signaling from macrophages (sending cell type) to fibroblasts (receiving cell type). Colour bar represents combined mean expressions of the receptor and ligand within the interacting cell types. Dot size and percentage represents the percentage of cells that express the interacting receptor-ligand pair, within that interacting cell type pair. Only receptors and ligands expressed in more than 10% of cells in the specific group were considered significant. Dotplots showing expression of the 7 macrophage receptors in the top 20 receptor-ligand pairs from **C)**, across the **D)** cell types and **E)** macrophage subclusters. **F)** UMAP showing *Cd74* expression within macrophages. Colour bar represents log fold change expression in each cell/dot. Dotplots showing expression of **G)** *Cd74* in macrophages and **H)** *Mif* in fibroblasts, across timepoints. Dotplots: colour bar represents mean expression of the receptor across all cells in that cell type, subcluster or timepoint. Dot size and percentage represents the percentage of cells that express the gene in that group. (Figure created with BioRender.com).

Discussion

Differences in gene expression and distribution between batches, for example for *Wif1*, could be explained by the fact that the samples were taken from different ventricular regions. Reports indicate that there are no major differences in fibroblast distribution within ventricular regions of the heart: LV, RV and IVS (Litviňuková *et al.*, 2020). However, considering the different functions of LV and RV, one would expect to see some differences. For certain cell types, differences in gene expression have been demonstrated between these regions. For example, recently *Pln* (Phospholamban) was found to be more highly expressed in the cardiomyocytes from the RV compared to the LV, in embryonic mouse hearts (de Soysa *et al.*, 2019). Future work might include comparison of gene expression in cells from individual scRNA-seq analyses of the separate ventricular regions.

As shown in the Venn diagram (Fig. 3. G), the logreg method for calculating DEGs identifies more genes that are not captured by the other methods. However, the *Scanpy* package currently does not report p-values or log fold changes for logreg, which makes it difficult to distinguish significant genes. Internal filtering has been implemented in the *rank_genes_groups* function for all methods, which can filter out lowly expressed genes and increase the power for detecting differences in gene expression (Bourgon *et al.*, 2010). In this analysis, using the internal filter for log fold change expression with the logreg method often resulted in a decrease of reported differentially expressed genes for timepoint D07 (see dotplots in Fig. 7. I: D07, Fig. 8. C: D07). Likewise, with the Wilcoxon method, applying the filters (genes=500, filtered q-value=1E-5, minimum log fold change=1) resulted in a decrease of reported DEGs. In particular, the F5 fibroblast subcluster only returned 10 DEGs. Whilst the lack of reported DEGs may be due to overall low abundance of significant cells and genes in

these specific groups, this could also be indicative of the presence of too many clusters, in which case, reducing the number of clusters might enable detection of a stable number of DEGs across all subclusters. Another contribution could be the step at which the filtering was applied. For example, recent analysis of the performance of certain scRNA-seq DEG calculation methods showed an overall decrease in the number of non-reported genes when genes were pre-filtered, compared to when genes were internally filtered (Soneson & Robinson, 2018). Although Wilcoxon was demonstrated to have robust performance, logreg was not analysed in this study. Therefore, this analysis could also benefit from exploring the effect of applying different thresholds of expression in pre-filtering and comparing DEGs from other multivariate models such as the Characteristic Direction (Clark *et al.*, 2014).

Another limitation of this analysis was that the induced experimental MI was performed by permanent coronary artery ligation, to match the conditions for grafting of cells used in the previous fate mapping study by the team (Nosedá *et al.*, 2015). As a disease model, it is easy to develop and reproduce compared to ischemia with reperfusion, but the latter is a better simulation of myocardial infarction as it currently exists. This is because most patients undergo urgent reperfusion therapy within the first few hours of MI. Ischemia reperfusion models tend to have less severe injuries which may cause differences in the gene expressions and temporal dynamics of cardiac stromal cells. However, this might introduce other difficulties, for example in collecting enough samples to be able to detect these smaller differences. As well as reperfusion models, the neonatal mouse heart model might tell us a lot about the underlying mechanisms of fibrosis and regeneration of mammalian hearts, and it might be interesting to compare scRNA-seq analyses of these 2 separate mouse models alongside this MI model.

In addition, previous single cell qRT-PCR analysis by the Nosedá and Furtado teams showed high expression of cardiogenic genes *Tbx20*, *Hand2*, *Gata4* and *Gata6* in cardiac fibroblasts (Nosedá *et al.*, 2015; Furtado *et al.*, 2014). The lack of cardiogenic gene expression in this analysis could be explained by low abundance of RNA coding for transcription factors, combined with insufficient sensitivity of the 10X Genomics platform.

Interestingly, the Nosedá team also found that CD31- PDGFRa+ Sca1+ cells are unable to spontaneously differentiate into cardiomyocytes. They previously confirmed that the proteins were expressed and appropriately nuclear-localized and concluded that it is not a disconnect between RNA and protein. Instead, this could be due to the lack of the factors' co-expression at the single cell level (Nosedá *et al.*, 2015). Also, it could be due to the lack of Myocardin (MYOCD) and Tbx5, which have been demonstrated to be required to drive CSC towards a cardiac muscle fate (Belian *et al.*, 2015). Also, in this analysis cardiomyocytes could not be directly analysed as the data was single cell and not single nuclei. Therefore, future research could investigate the expression of Myocardin across fibroblast subclusters and across timepoints at a single cell level, and also have a focus on trajectories or RNA velocity of cell types to identify the direction of potential differentiation of cardiac stromal cells and take advantage of the PDGFRa fate mapping, which could be used to identify the embryonic layer from which these cardiac stem progenitors originate. However, one point to note is that RNA velocity relies on splicing information which is not present in single nuclei data.

Using *CellPhoneDB* analysis, we demonstrated that the interaction between macrophage-expressed receptor CD74 and fibroblast-expressed ligand MIF was predicted as one of the most significant interactions between fibroblasts and

macrophages. The functions of MIF are not yet well known and experimental research show conflicting results. Studies investigating the general role of MIF have shown that it supports pro-inflammatory processes and more specifically, promotes post-infarction cardiac rupture and infiltration of leukocytes by binding to chemokine receptors CXCR2 and CXCR4 post-MI, which are critical immune cells for ECM remodelling (White *et al.*, 2014; Berghagen *et al.*, 2007). However, it has also been shown to mediate cardioprotective effects in response to ischemia reperfusion injury by inhibiting the c-Jun N-terminal kinase (JNK) pathway in ischemia reperfusion injury when bound to the CD74 receptor (Qi *et al.*, 2009). CD74 Recently, investigation into the function of the soluble CD74 (sCD74) receptor ectodomain and its interaction with circulating MIF revealed that this interaction induces RIP1/3 (receptor-interacting serine/threonine-protein kinase 1/3) dependent necroptosis in myofibroblasts derived from cardiac fibroblasts (Soppert *et al.*, 2018). Further, sCD74/MIF-induced cell death was only demonstrated in myofibroblasts. Preliminary clinical data also indicated that patients with heart failure had a lowered sCD74 to MIF ratio (Wu *et al.*, 2016), a characteristic that could also be explored in scRNA-seq analyses of murine hearts post-MI. This makes the CD74/MIF interaction a promising target for further investigation into the differences in response to sCD74/MIF, its dynamic response and its role in regulating immune cells and myofibroblasts following MI. Following on from this analysis, it would be interesting to see if there are differences in expression of this interaction across timepoints and if so, this could indicate specific roles of this receptor-ligand pair in response to MI.

Overall, the results of this analysis give us a deeper insight into the heterogeneity of cardiac stromal fibroblasts, their functions post-MI, their interactions with macrophages and the characterisation of SP cells. Many concepts still remain unclear,

such as the turnover rate of cardiomyocytes in mammalian adults, whether a true cardiac stem cell exists in the heart and the overlapping signatures of cardiac stem progenitors with fibroblasts.

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